

Project CP-05

RetailIQ

Demand Forecasting and Multi-Tool Business Assistant

Assigned to	Batch 5, Room 306, Block 3
Cohort	52 students, MCA (AIML), IT, MCA and M.Sc
Team structure	11 teams
Domain	Retail, Supply Chain and Enterprise AI
Submission	Day 12, 15 September 2026

1. Background

Forecasting error costs money at both ends. Stockouts lose the sale; overstock ties up capital and ends in markdown. The analytics usually exist, but they sit in fixed dashboards, so the only questions that get answered are the ones someone built a report for in advance.

2. Scope of Work

Build a demand forecasting platform paired with a business assistant that answers natural-language questions by selecting among a SQL analytics tool, a forecasting tool and a retrieval tool over internal documentation. The tool-selection architecture is the defining requirement of this project.

Most of the components listed in Section 5 correspond to work already completed during the programme. The task now is to consolidate that work into one coherent application, close the gaps, and produce something that runs end to end and can be defended under questioning.

3. Team Structure

The room is divided into 11 teams. Each team builds one complete instance of the project on the same dataset, which allows the work to be supported and compared on common ground. Responsibilities below may be shared in teams of three.

Area	Responsibility
Data	Acquisition, cleaning, pipeline and schema. Accountable for data quality.
Modelling	Feature engineering, training, validation method and evaluation reporting.
AI layer	Prompt design, retrieval pipeline, tool calling and guardrails.
Application	Interface, deployment, repository and documentation.
Integration	Version control, merging, conflict resolution and keeping the build runnable.

Each student must be able to explain every part of the submission, including work they did not personally write. Individual questioning forms part of the assessment.

4. Data

Primary	Retail transactions, minimum 100,000 rows across multiple stores, products and categories, spanning at least twenty-four months, with date, quantity, price, promotion flag and store attributes.
Secondary	Product master, store master, inventory levels and promotional calendar.
Knowledge base	Internal documentation covering inventory policy, markdown rules, supplier terms, returns policy and

	standard operating procedures.
Source	Kaggle retail datasets, Walmart or Rossmann store sales data, or the dataset pack issued with this brief.
Restriction	Public or synthetic commercial data only. No real supplier contracts, pricing agreements or customer records.

5. Required Components

Each item below is a requirement and is assessed. Where a component was built earlier in the programme, it must be integrated into the application rather than submitted separately.

Component	Requirement
Data preparation	Reproducible pipeline across multiple source files. Reconcile product and store masters against the transaction table. Handle missing promotion flags, negative quantities from returns, and stores opening or closing mid-period.
Analysis	Examine seasonality, promotional lift and price sensitivity. Test whether promotional periods differ materially in volume. Quantify the contribution of leading products and stores.
Database	Star schema with a fact table and dimension tables. This is the analytical layer the assistant will query, so it must support sales by any dimension, period-over-period growth, top-N per group, running totals and inventory cover.
Dashboard	Executive view covering revenue and volume, category performance, store ranking, promotional effectiveness and stock position, with drill-through to product level.
Forecasting model	Forecast at product-store-week level. Engineer lag features, rolling statistics, promotional and calendar features. Use forward-chained validation. Report MAPE and RMSE, and translate the error into commercial terms.
Sequence model	Implement a sequence model and compare it against the tree-based approach. Derive category features from product description text and test whether they improve accuracy.
Agent layer	Planner-executor agent exposing at least three tools: a SQL tool translating business questions into queries against the star schema, a forecast tool invoking the trained model for a given product and horizon, and a retrieval tool over internal documentation. The agent selects the appropriate tool or sequence, executes, and returns a single grounded answer with its reasoning visible.
Application	Interface with executive dashboard, forecast explorer with horizon selection, and the conversational assistant showing its tool-selection reasoning.

6. Functionality

The deployed application must demonstrate the following. Anything not working at the time of presentation counts as not delivered.

- Forecast at product-store-week level with stated error metrics
- Executive dashboard covering commercial indicators
- Agent selecting autonomously between SQL, forecasting and retrieval
- Visible reasoning trace showing tool selection
- Answers cited to the source document where policy is involved
- Natural-language interface requiring no SQL knowledge from the user

7. Submission

Item	Requirement
Repository	Public repository containing all code, notebooks, SQL files and documentation, with a commit history that reflects the work.

Item	Requirement
README	Problem statement, approach, stack, architecture diagram, screenshots, results with metrics, setup instructions and the live link.
Application	Publicly accessible deployment. It must be running at the time of presentation.
Notebooks	Analysis, modelling and evaluation notebooks, executed with outputs visible.
SQL	Schema and query files, both runnable against the supplied data.
Dashboard	Source file and exported PDF.
Slides	Eight to ten slides covering problem, data, approach, demonstration, results and limitations.

8. Assessment

Marked out of 100. The result contributes to the projects component of the placement readiness score.

Criterion	Marks
Data preparation and analysis	15
Database design and querying	10
Dashboard	10
Modelling and validation	20
Retrieval and assistant layer	20
Deployment and documentation	10
Presentation and defence	15

9. Optional Extensions

Attempt these only once everything in Section 6 is working and deployed. Extensions are noted in the presentation but do not compensate for an incomplete core build.

- Inventory optimisation deriving reorder points from the forecast
- Supervisor and worker agents with specialised sub-agents
- Scheduled reporting generated and distributed by the agent
- Model Context Protocol server exposing the tools through a standard interface

Queries on this brief should be raised with your batch trainer or the programme coordinator.

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